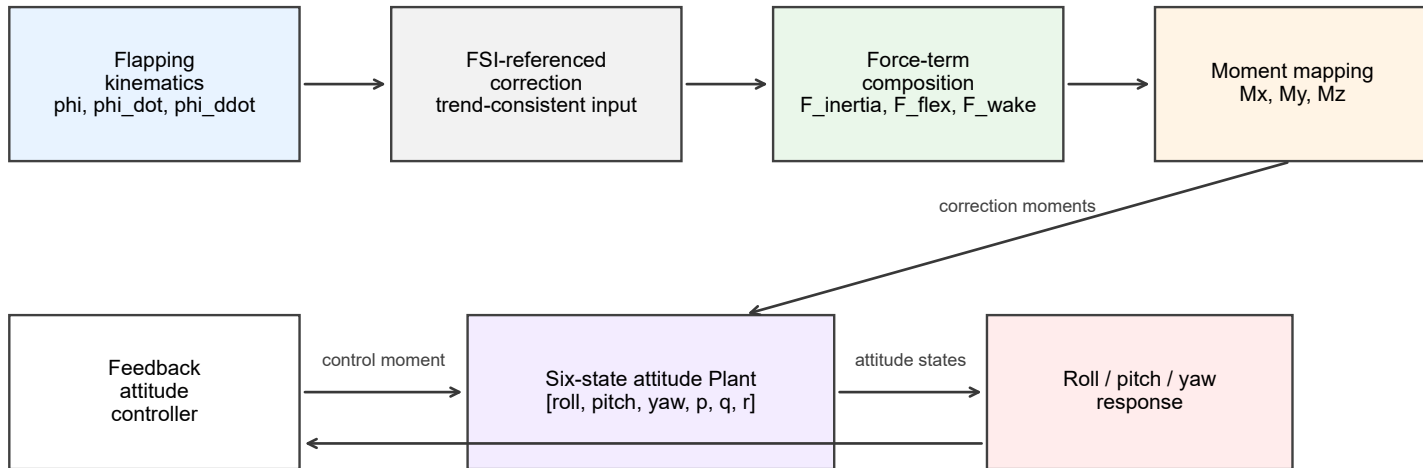




$$J\dot{\omega} = M_c + M_{inertia} + M_{flex} + M_{wake} - D\omega$$

Simplified mechanism-based Simulink model; FSI-referenced indicates trend guidance, not high-fidelity FSI validation.